11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

1.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer's unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05's adjustment tools are powerless here.

1.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs three things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; $F < 10$ is a "weak instrument" danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).

1.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the "compliers,"* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

> **On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement / intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument.

1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved `engagement` drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized `encouragement` (the instrument) nudges exposure without touching sales directly.

```
TRUE exposure->sales effect €15  ·  first-stage: exposure 56%->77% (shift +21
pp), F = 156

   encouragement  ad_exposure      sales
0            1.0          0.0  63.748992
1            1.0          1.0  64.636639
2            1.0          1.0  46.507627
3            0.0          1.0  75.599770
4            0.0          1.0  70.577754
```

1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ($\mathrm{Cov}(\text{exposure}, \varepsilon) \neq 0$), so OLS is biased. A valid instrument Z (encouragement) needs three things:

- **Relevance** — Z moves exposure. *Testable* (the first-stage F above; F<10 = weak).
- **Exclusion** — Z affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** — Z independent of the confounder. Holds by randomization.

What IV identifies — LATE (Imbens–Angrist): under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.

- **Reduced form** — regress sales on the *instrument* (encouragement -> sales). This captures only the effect that flows through the clean, randomized channel.
- **First stage** — regress exposure on the instrument (encouragement -> exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form ÷ first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity ρ rather than assuming it away).

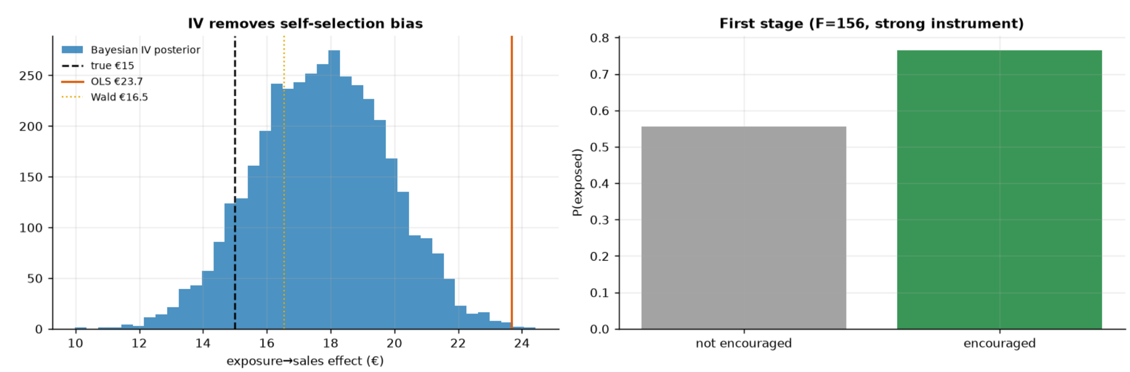
```
OLS (biased)      €23.7
Reduced form (Z->Y) €3.5  ÷  first stage (Z->D) 0.21  =  Wald/IV ratio €16.5
Bayesian IV        €17.7  (true €15)  90% credible interval [€14.3, €21.1]
IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0
Estimated endogeneity  rho(exposure err, sales err) = +0.22 [90% +0.10, +0.33] -
nonzero rho is why OLS (€23.7) sits above IV (€17.7).
```

1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior's 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy's default 2SLS-centred priors would produce.)

One honest note on LATE (and who the compliers are). IV recovers the effect for *compliers* — the customers whose exposure the encouragement actually moves — not the whole population. Under monotonicity that group is exactly the **first-stage shift: +21 pp (56% -> 77%)**, the estimated complier share. In *this* simulation every customer's true effect is €15 (the effect is homogeneous), so **LATE = ATE** and grading the IV against €15 is a fair test. On real data, where the effect varies across customers, IV would recover the complier-weighted LATE, which generally **!= the ATE** — and that is the right quantity for a budget call, since it answers *"what does an extra exposure do for the customers we can actually move?"*

```
self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)
```

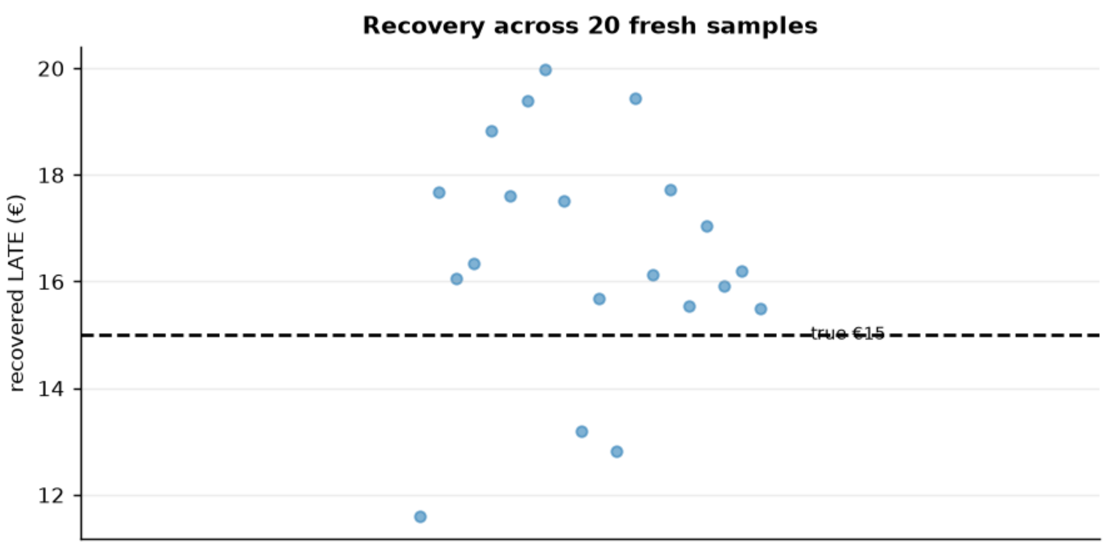


How to read this. *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a little finite-sample noise (a euro or two above 15), but its posterior — now honestly wide, thanks to weakly-informative priors — comfortably **covers €15**; the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of "effect" that were never causal. If you budgeted on OLS you'd over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** (F » 10). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

1.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each) and check both.

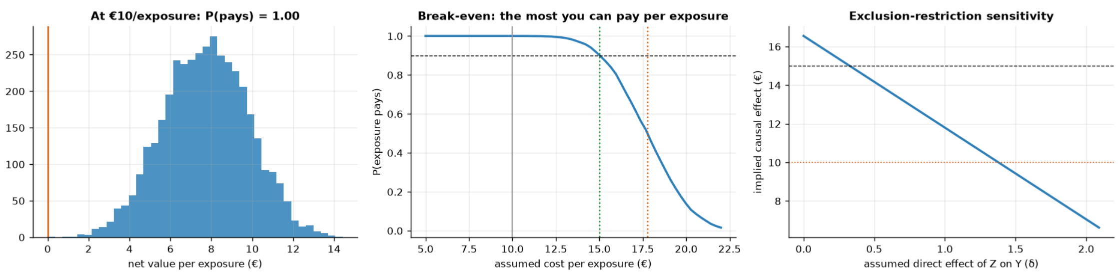
```
IV across 20 seeds: mean €16.5 (true €15)  bias +1.5  sd €2.1  ·  90% interval
covers truth in 16/20 seeds.
The single committed seed (37) lands high (€18), but the estimator is UNBIASED
on average - the multi-seed view is the
honest calibration check, and it is why we trust the ranking/verdict, not one
draw.
```



1.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

```
At €10/exposure: causal net €7.7   ·  P(pays) 1.00 -> BUY exposure (to the
complier margin)
  P(pays)=1.00 is saturated, not assumed: the whole posterior [€14.3, €21.1]
sits above €10.
Break-even sweep: exposure clears the 90%-confidence bar up to €15.0/exposure
(the budget cap); at €17.8 it is a coin flip.
  So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight.
Exclusion stress: the BUY call survives unless the instrument's direct effect
exceeds delta~1.4 (out of a €3.5 reduced form) -
  trusting OLS (€24) would overstate the value and overspend.
```



1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.